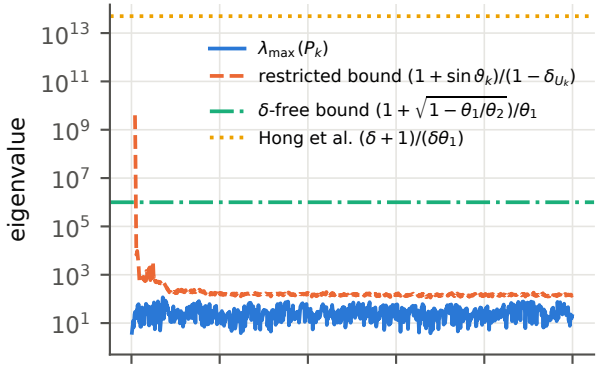
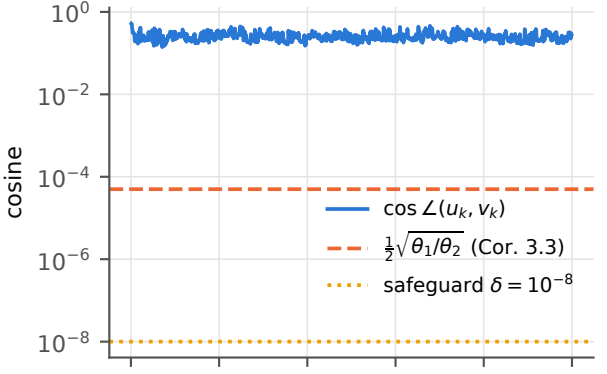

eigenvalue
iteration k
$\lambda_{\max}(P_k)$
restricted bound $(1+\sin\vartheta_k)/(1-\delta_{U_k})$
δ-free bound $(1+\sqrt{1-\theta_1/\theta_2})/\theta_1$
Hong et al. $(\delta+1)/(\delta\theta_1)$
10^{13}
10^{11}
10^{9}
10^{7}
10^{5}
10^{3}
10^{1}
0
200
400
600
800
1000


cosine
iteration k
$\cos\angle(u_k, v_k)$
$\frac{1}{2}\sqrt{\theta_1/\theta_2}$ (Cor. 3.3)
safeguard $\delta = 10^{-8}$
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
0
200
400
600
800
1000